

# VLHIVE 151 DIGITAL FX

Date:

Type:

Firm Name:

Project:

The VLHive 151 Digital FX is a true evolution in entertainment lighting, a next-generation luminaire with advanced digital animation effects never before possible with an entertainment lighting fixture. Designed and built in Vari-Lite's Dallas, TX, USA headquarters, the VLHive includes a digital light engine, with 45,000 lumens of output and pixel-level control. Thanks to user-loadable effects and familiar DMX control, the luminaire fosters better collaboration between lighting, video, and creative teams while remaining, at its heart, a light fixture. Built with Vari-Lite's commitment to longevity and featuring the widest range of design possibilities of any fixture on the market, the VLHive 151 promises to provide fresh design looks for years to come.



- Patent-pending multichromatic light engine featuring 151 individually controlled LEDs with calibrated RGBL color and a max output of 45,000 lumens- create bright, digitally generated images and animation effects
- High-impact 300mm (12") front lens fitted with patent pending Vari\*glass projection technology- large impact washes, digitally created beams, and direct-view set piece effects
- Digital images (80 pre-installed and 60 user-installable\*\*) with scaling, keystoneing, polar positioning, and more- endless variety of "gobo" type images without mechanical constraints  
\*\*Requires Vari-Lite HiveCreator software to upload
- Digital animation effects (50 pre-installed and 50 user-installable\*\*)- produce video-style animation effects where the output remains the true light source and where black is true black (out), not "video black".
- Integrated hue, saturation and contrast control of animation- fine tune animations for unlimited possibilities in creating digitally driven looks from the lighting desk
- 16-bit dimming control, continuously rotating digital 4 blade framing shutters, and 100% closing digital iris- global master-level control of the projected beam with fewer physical optics for max output
- Patent pending SmartColor+ system, optional open-source color mixing, calibrated VL library preset colors, background colors and color effects- advanced color system that expands color control options
- Hybrid optical system combining traditional 2:1 optical and digital scale- produces wide angle washes and effect or pinpoint beams
- Multiple control options, including DMX (RDM) via 5 PIN XLR, Ethernet protocols and video integration such as NDI\*\*\*connection via dual Ethercon control connection- comprehensive digital connectivity for every application  
\*\*\*Function to be added in future release.
- Robust IP65 mechanical design coupled with integrated advanced feature sensors and data log reporting- Built to tour

For detailed product information, please refer to the Product Guide at <https://www.vari-lite.com/global/products/vlhive-151-dfx>



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# Specifications

Due to continuous improvements and innovations, specifications may change without notice.

## Source

|                              |                                                        |
|------------------------------|--------------------------------------------------------|
| LED System                   | Multichromatic Light Engine                            |
| Source Max Power Consumption | 1500 W                                                 |
| Source Lumens                | 110,000                                                |
| LED Refresh Rate             | 250hz - 2000hz                                         |
| Dimming                      | Electronic 16-bit                                      |
| Dimmer Curves                | Square Law (Default), Linear, S -Curve                 |
| Dimmer Control Option        | Dimmer Snap On/Off<br>On reduces auto dimmer smoothing |
| Strobe Speed                 | 0.5hz to 30hz                                          |
| Strobe Modes                 | Continuous, Random, Random Sync                        |

## Optical Performance

|                                                                                                                                                                                                                                                                                                                        |                                                                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| Len Size                                                                                                                                                                                                                                                                                                               | 300 mm (12.0 in)                                               |
| Optical Zoom                                                                                                                                                                                                                                                                                                           | 2:1 Ratio                                                      |
| Digital Zoom                                                                                                                                                                                                                                                                                                           |                                                                |
| Pixel Ring 1 = 0.6°-1.3° - 1 Pixel; Pixel Ring 2 = 3.8°-9.7° - 7 Pixels; Pixel Ring 3 = 8.2°-17.9° - 19 Pixels; Pixel Ring 4 = 12.5°-26° - 37 Pixels; Pixel Ring 5 = 16.9°-33.8° - 61 Pixels; Pixel Ring 6 = 21.2°-41.3° - 91 Pixels; Pixel Ring 7 = 25.4°-48.4° - 127 Pixels; Pixel Ring 8 = 26.4°-50.1° - 151 Pixels |                                                                |
| Optical Zoom Control Digital Zoom Control                                                                                                                                                                                                                                                                              | Motorized 8-Bit Digital 8-Bit                                  |
| Image definition control (Focus)                                                                                                                                                                                                                                                                                       | Motorized 8-Bit Control                                        |
| Output Lumens                                                                                                                                                                                                                                                                                                          | Standard - 45,000<br>Studio - NEED DATA<br>Whisper - NEED DATA |
| CCT Range Smart Color Mode                                                                                                                                                                                                                                                                                             | 2700K - 10,000K                                                |
| CRI RA                                                                                                                                                                                                                                                                                                                 |                                                                |
| Smart Color 3200k - 88 Smart Color 4000k - 88 Smart Color 6500k - 82 Smart color 10,000k - 74 ***Range Adjusted via CRI enhancement color channel to boost CRI 'v' MAX output                                                                                                                                          |                                                                |
| TM30                                                                                                                                                                                                                                                                                                                   |                                                                |
| Smart Color 3200k - NEED DATA RANGE Smart Color 4000k - NEED DATA RANGE Smart Color 6500k - Need DATA RANGE Smart color 10,000k - NEED DATA RANGE                                                                                                                                                                      |                                                                |
| CSQ                                                                                                                                                                                                                                                                                                                    |                                                                |
| Smart Color 3200k - NEED DATA RANGE Smart Color 4000k - NEED DATA RANGE Smart Color 6500k - Need DATA RANGE Smart color 10,000k - NEED DATA RANGE                                                                                                                                                                      |                                                                |
| TLCI                                                                                                                                                                                                                                                                                                                   |                                                                |
| Smart Color 3200K - 92 Smart Color 4000K - 89 Smart Color 6500K 82 Smart Color 10,000K - 55                                                                                                                                                                                                                            |                                                                |

## Color System

|                             |                                                                                     |
|-----------------------------|-------------------------------------------------------------------------------------|
| Additive color mixing       | RGBL                                                                                |
| Smartcolor+ Open Source     | 16-Bit digital control                                                              |
| Calor calibration           | Yes                                                                                 |
| Color Mixing Control Option | Color Snap On/Off<br>On reduces auto smoothing color to create faster color changes |
| Smartcolor+                 | 2700 K to 10,000 K                                                                  |
| Color Presets Primary       | 33 Preset Color + 20 User recordable Colors                                         |
| Background Color            | 33 Color Preset + Invert Color                                                      |
| Image Color Effects         | 40 Image applied color effects                                                      |
| Animation Color Control     | Hue, Saturation and Contrast Control                                                |

## Imaging

|                 |                                                 |
|-----------------|-------------------------------------------------|
| Digital Framing | 4 Blade - Full Wipe<br>360° continuous rotation |
| Digital Iris    | 100% closing                                    |

## Electrical

|                    |                                                                        |
|--------------------|------------------------------------------------------------------------|
| Max output Power   | 2400 W (Inc PFC) Smart Power management limits max                     |
| Consumption        | current draw based on supply voltage monitoring                        |
| Power Connection   | TrueOne Compatible<br>*Power cable not included - input connector only |
| Input Voltage      | 100-250VAC 50/60 Hz                                                    |
| Standby Power Draw | NEED DATA                                                              |

## Data

|                     |                                                           |
|---------------------|-----------------------------------------------------------|
| Supported Protocols | DMX (RDM), ArtNet, sACN, RDM NET, NDI* *Later 2025        |
| DMX Modes           |                                                           |
| Smart Color Mode    | 85 Channels                                               |
| Open Source Mode    | 85 Channels                                               |
| Data Connections    |                                                           |
| Data Connections    | 5-pin XLR, 4 x RJ45, Wireless DMX (Lumen Radio), USB Data |

## Noise Level / Fan Control (Base Level 30 dB)

|                                                                                                                              |                                |
|------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| Standard Mode                                                                                                                | NEED DATA                      |
| Studio Mode                                                                                                                  | NEED DATA                      |
| Whisper Mode                                                                                                                 | NEED DATA                      |
| Fan Mode                                                                                                                     | Continuous (On) / Auto Ramping |
| Fan Level Preset                                                                                                             | Standard, Studio, Whisper      |
| Cooling System Type                                                                                                          |                                |
| Patent Pending, fully intergrated adavnce engine and cooling combination heat transfer system with active forced air support |                                |

## Movement

|                    |                     |
|--------------------|---------------------|
| Pan Angle          | 540°                |
| Pan Speed          | NEED DATA           |
| Tilt Angle         | 240°                |
| Tilt Speed         | NEED DATA           |
| Calibration System | V*Track Calibration |

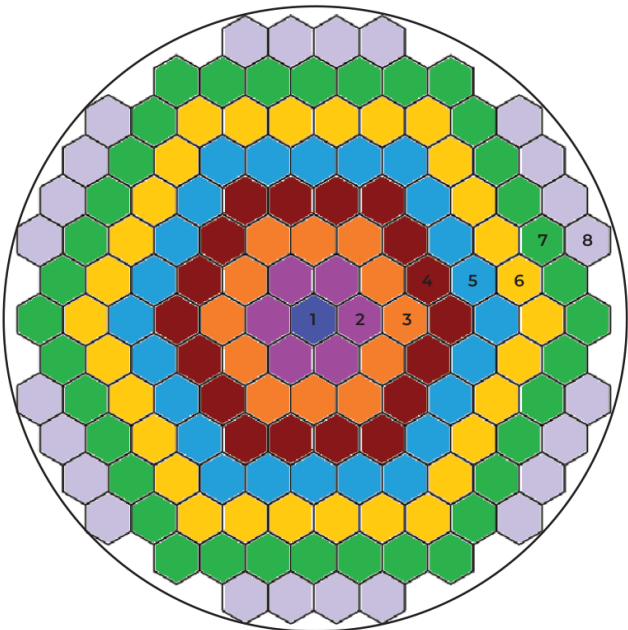
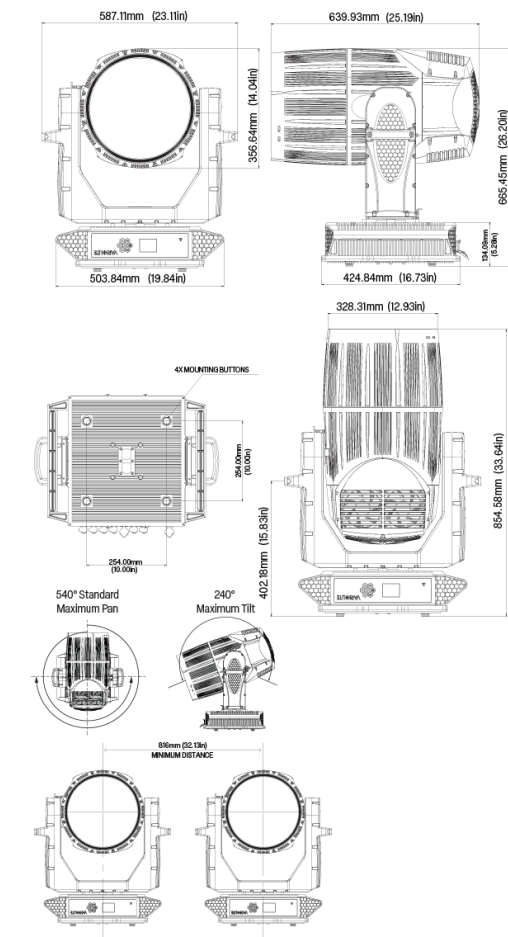
## Mounting

|                              |                                                                                      |
|------------------------------|--------------------------------------------------------------------------------------|
| Hanging Type                 | 2 x VL Hook Rails L                                                                  |
| Hanging Centers              | 816.0 mm (32.1 in)                                                                   |
| Mounting/Hanging Orientation | Hanging / Floor - Side<br>*See user manual for side hang guidance and restrictions   |
| Secondary Safety Wire Point  | 1 x Safety point located on underside of upper enclosure<br>Safety wire not included |

## Photometrics and IES Files

|                                                                                                                                                                                                                                               |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| Narrow Optical @10M / 32.8ft Pixel Ring 1 = 1 Pixel Pixel Ring 2 = 7 Pixels Pixel Ring 3 = Pixels Pixel Ring 4 = 37 Pixels Pixel Ring 5 = 61 Pixels Pixel Ring 6 = 91 Pixels Pixel Ring 7 = 127 Pixels Pixel Ring 8 = 151 Pixels lux (0 fc)   |  |
| Narrow Optical @20M / 65.62 ft Pixel Ring 1 = 1 Pixel Pixel Ring 2 = 7 Pixels Pixel Ring 3 = Pixels Pixel Ring 4 = 37 Pixels Pixel Ring 5 = 61 Pixels Pixel Ring 6 = 91 Pixels Pixel Ring 7 = 127 Pixels Pixel Ring 8 = 151 Pixels lux (0 fc) |  |
| Wide Optical @10M / 32.8ft Pixel Ring 1 = 1 Pixel Pixel Ring 2 = 7 Pixels Pixel Ring 3 = Pixels Pixel Ring 4 = 37 Pixels Pixel Ring 5 = 61 Pixels Pixel Ring 6 = 91 Pixels Pixel Ring 7 = 127 Pixels Pixel Ring 8 = 151 Pixels lux (0 fc)     |  |
| Wide Optical @20M / 65.62 ft Pixel Ring 1 = 1 Pixel Pixel Ring 2 = 7 Pixels Pixel Ring 3 = Pixels Pixel Ring 4 = 37 Pixels Pixel Ring 5 = 61 Pixels Pixel Ring 6 = 91 Pixels Pixel Ring 7 = 127 Pixels Pixel Ring 8 = 151 Pixels lux (0 fc)   |  |
| For IES files and photometric reports, please refer to the product page:<br><a href="https://www.vari-lite.com/global/products/vlhive-151-dfx">https://www.vari-lite.com/global/products/vlhive-151-dfx</a>                                   |  |

Dimensions



| Pixel Ring | Opt. Zoom Narrow | Opt. Zoom Wide | Total Pixels used |
|------------|------------------|----------------|-------------------|
| 1          | 0.6°             | 1.3°           | 1                 |
| 2          | 3.8°             | 9.7°           | 7                 |
| 3          | 8.2°             | 17.9°          | 19                |
| 4          | 12.5°            | 26.0°          | 37                |
| 5          | 16.9°            | 33.8°          | 61                |
| 6          | 21.2°            | 41.3°          | 91                |
| 7          | 25.4°            | 48.4°          | 127               |
| 8          | 26.4°            | 50.1°          | 151               |

Field Angle by Pixel Diameter

Physical

|                   |                                                                                              |
|-------------------|----------------------------------------------------------------------------------------------|
| Construction      | Superstructure made of Aluminum, magnesium and steel metal plates, Covers Die Cast Magnesium |
| IP Rating         | IP65                                                                                         |
| Standard Finish   | Black, White                                                                                 |
| Dimensions        | 587 x 855 x 425 mm (23.1 x 33.6 x 16.7 in)                                                   |
| Weight            | 48.5 kg (107.0 lb)                                                                           |
| Surface Treatment | a minimum of 2000 hours continuous for salt spray resistant finish                           |

Certification and Safety

|              |                             |
|--------------|-----------------------------|
| Approbations | CE,FCC,RCM,ISED,UKCA,ETL,CB |
|--------------|-----------------------------|

Gobo and Effect Wheels

|                               |    |
|-------------------------------|----|
| Factory Loaded Digital images | 80 |
| User Loaded Digital images    | 60 |
| Factory loaded animations     | 50 |
| User loaded animations        | 50 |

# Luminaire and Accessories

| Product type                                                                                                              | Order Code |
|---------------------------------------------------------------------------------------------------------------------------|------------|
| VLHIVE 151 DIGITAL FX, BLACK                                                                                              | 74820-001  |
| VLHIVE 151 DIGITAL FX, 1500W MULTICHROMATIC DIGITAL LIGHT ENGINE, C/W 2 X HANGING RAILS, AND POWER INPUT CONNECTOR, BLACK |            |
| VLHIVE 151 DIGITAL FX, WHITE                                                                                              | 74820-101  |
| VLHIVE 151 DIGITAL FX, 1500W MULTICHROMATIC DIGITAL LIGHT ENGINE, C/W 2 X HANGING RAILS, AND POWER INPUT CONNECTOR, WHITE |            |
| Accessories                                                                                                               |            |



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Vari-Lite  
<https://www.vari-lite.com/global/products/vlhive-151-dfx>